



Keysight Technologies Response to Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Keysight is a leading multinational technology provider. Keysight develops solutions that accelerate innovation to connect and secure the world. Keysight supports mission critical solutions with design, emulation, complete testing, manufacturing, optimization and network security monitoring in high growth industries.

Keysight is partnered with 10/10 top automotive companies, 90% of technology companies, 95% of top mobile operators, 25/25 top semiconductor companies, 90% top aerospace and defense contractors with over 30,000 top customers in over 100 countries.

With its vast knowledge of industries, markets, customers and over 15,000 employees and over 2,000 patents/pending patent applications, Keysight's experience informs its business strategy to drive customer success.

AI is a transformative force with the potential to address complex business and global challenges. Ensuring the continued advancement of AI capabilities is essential to U.S. economic growth and national security.

This document makes recommendations below in response to "Request for Information on the Development of an Artificial Intelligence (AI) Action Plan.

- **Facilitate Rapid AI Adoption** – Ensure policies encourage LLM/AI model development with scalable testing and minimal restrictions.
- **Strengthen AI Infrastructure** – Validate cloud and on-prem AI systems for high-usage reliability and secure deployment.
- **Enhance AI Security & National Defense** – Monitor AI access to sensitive data, classified systems, and operational networks to prevent security threats.
- **Implement Robust AI Risk Mitigation** – Establish safeguards against data exfiltration, model poisoning, and unauthorized AI use in government and private sectors.

Focus Areas

1. Encouraging AI Adoption

- a. U.S. companies are rapidly integrating LLMs, ML, SLMs, and agentic AI technologies to enhance efficiency and maintain global leadership.
- b. Policies should facilitate the rapid development of internal LLMs and AI applications, incorporating scalable testing of user workflows.

2. Enhancing AI Infrastructure for Scalable Deployment

- a. Policies should support the validation of cloud and on-premises AI infrastructure to ensure reliability under high-usage conditions.
- b. AI models must be production-ready with confidence in their performance, security, and efficiency.

3. Strengthening AI Security & National Defense

- a. AI is being weaponized to exploit vulnerabilities, automate cyberattacks, and undermine existing security frameworks. This requires improvements in automated responses given the speed of these AI augmented cyber weapon systems.
- b. AI strengthens cybersecurity by enhancing threat detection, decision-making, and risk identification. AI-driven cybersecurity solutions should be leveraged to protect national interests while mitigating adversarial AI threats.
- c. Policies should support comprehensive monitoring of AI use across networks, particularly where access to sensitive data, classified information, and operational or weapons systems could pose national security risks.

4. Mitigating AI-Driven Risks & Ensuring Secure AI Use

- a. Both controlled and open-source AI models are susceptible to manipulation, data extraction, and intellectual property theft.
- b. AI systems must include safeguards to prevent unauthorized access and leakage of confidential information.
- c. Policies should ensure seamless monitoring and visibility into AI use within government agencies and the private sector.

To maintain U.S. leadership in AI, policies should prioritize innovation, security, and responsible development while ensuring AI infrastructure and applications are scalable, secure, and resistant to adversarial threats. Additionally, AI security policies must safeguard national security interests by monitoring AI's role in accessing sensitive data and operational systems.

Recommended AI Actions

1. Continuous AI Application Development

To support key innovations and adoption of AI in the public and private sector, it is critical that rapid progress is made with frontier models of AI. Being able to develop, validate, and continue the iterative process is critical and necessary for the US to keep and gain on its competitive advantage.

It is critical to optimize the speed of creating and automating test-case creation to cover various AI first use-case journeys for an improved UX flow, finding defects and ultimately improving time to market.

Policies should be put in place that assist in the rapid development of internal LLM or AI model applications with an integrated process to rapidly test the user workflow at scale.

- **Leverage AI & Digital Twin Testing** – Enhance test coverage, user experience, and QA efficiency.
- **Automate Cross-Platform Testing** – Ensure compatibility across all devices and platforms.
- **Validate AI Application Interfaces** – Monitor performance (TPS, CPS) and detect UI errors.
- **Optimize for Speed & Security** – Use a fast, secure, and user-friendly testing solution.

2. AI Infrastructure Validation And Security Hardening

To support high-performance AI frontier models and applications, advanced infrastructure is a key competitive advantage. Cutting-edge data center technologies—including compute, DPU, GPU, storage, and networking—are essential. Optimizing costs, efficiency, and operations enables the U.S. private and public sectors to drive innovation while minimizing operational expenses.

Continuously monitor, assess, and validate the performance and security posture of the AI infrastructure by implementing automated testing, and rigorous security audits. Use high performance benchmark solutions to gain efficiency, with regular checks on security protocols to mitigate emerging threats.

Policies should be put in place that assist in validating Cloud and on-premises AI infrastructure that can handle high usage and be brought into production with confidence.

- **Deploy Flexible Traffic Agents** – Use containers, VMs, Kubernetes, or cloud (AWS, GCP, Azure) for scalable testing.
- **Simulate Real AI & Application Workloads** – Validate performance under real-world conditions.

- **Test Training Workload Performance** – Measure TPS, CPS, throughput, and IP scale.
- **Assess Security & Attack Mitigation** – Simulate cyber threats and malicious AI prompts to strengthen defenses.

3. Monitoring AI Operations

Network Visibility is crucial for maintaining operational excellence, optimizing network, application and user experience, ensuring compliance of data at rest and data in motion and importantly, maintaining security against nation-state and cyberattacks.

With the rise of ransomware and AI-generated advanced persistent threats (APT), network visibility has an ever-increasing importance in delivering enhanced security, compliance, incidence response, deep forensics and for operational insights.

There is a need to promote collaboration between regulatory agencies, private sector cybersecurity professionals, AI infrastructure operators and AI architects and developers to help establish safeguards to mitigate evolving AI-based risks and AI-targeted attacks.

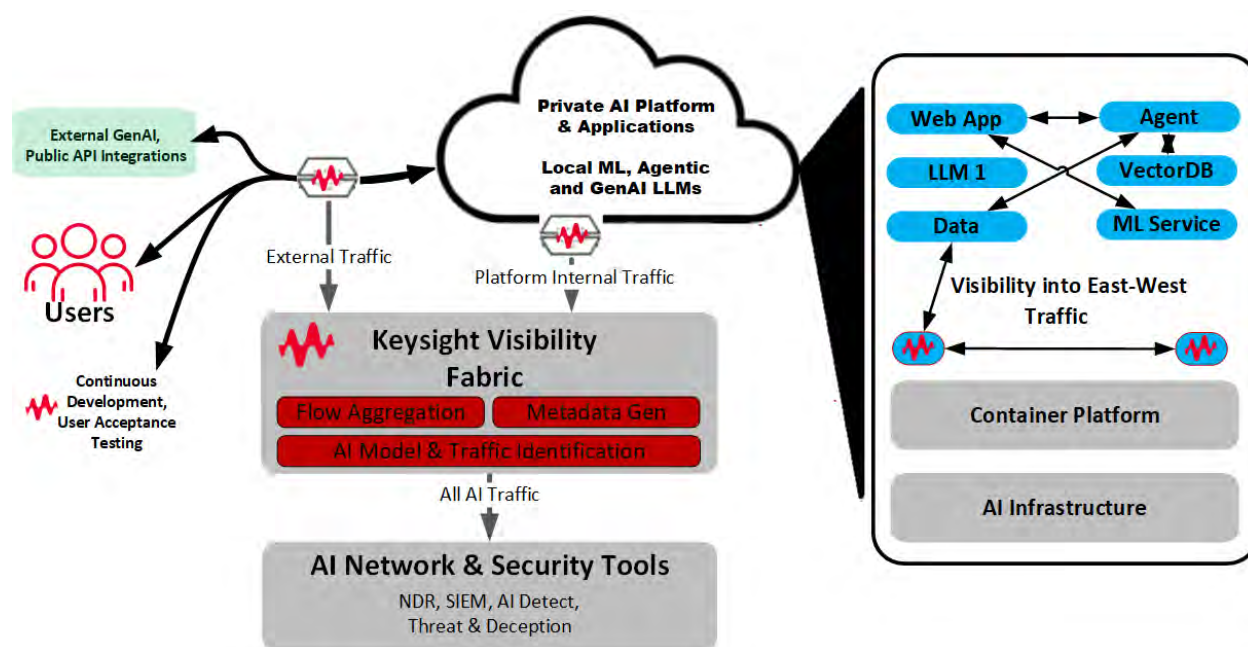
Policies should be put in place that allow seamless network monitoring and visibility into the use of AI in government agencies and private sector to prevent malicious use, including data exfiltration or model poisoning or ways to circumvent guardrails on AI to perform tasks that would otherwise not be authorized.

- **Enable Visibility into the Use of AI** – Monitor critical network segments with unauthorized use of AI or use of AI in unintended domains.
- **Leverage Deep Packet Inspection (DPI)** – Ensure detailed data analysis for detecting and filtering GenAI applications.
- **Enhance Detection Across Traffic Types** – Apply filtering to both encrypted and cleartext traffic, with advanced options for cleartext.
- **Implement Decryption Capabilities** – Gain visibility into GenAI application layers, crucial for analyzing encrypted traffic.
- **Use Session-Aware Detection** – Gain better accuracy over traditional packet-based filtering.
- **Perform Deep Packet Inspection (DPI)** – Analyze multiple protocol headers and payloads, including encrypted traffic (SSL, TLS, QUIC).
- **Leverage Application Signatures** – Enable high-performance real-time detection and filtering.
- **Utilize Dynamic & Static Signatures** – Apply both for flexible filtering and metadata generation.

Sample Architecture of AI Development, Infrastructure and Monitoring

This includes continuous development, user acceptance testing for private AI platforms and applications. Further, the ability to stress test complete AI infrastructure such as compute, storage and networking I/O capabilities in the cloud or on-premises.

Operational network visibility is required to monitor the use of generative AI, use of Agentic AI for deep network and business automations and implementation of defense in depth in support of maintaining security threats monitoring to detect and respond in real-time.



Remarks and Recommendation

Implementing these recommendations will not only strengthen the U.S. AI ecosystem, but it will also drive significant economic benefits.

By accelerating AI development, optimizing AI infrastructure, and enhancing security for AI applications through network visibility, both the public and private sectors will benefit from leading-edge innovations while reducing operational costs.

A streamlined, secure, and high-performing AI landscape will attract investments, enhance global competitiveness, and advance new market opportunities.

Furthermore, proactive policies that support AI-driven advancements can promote economic growth, ensuring that the U.S. remains at the forefront of the AI revolution in the new digital economy.

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